

Second Terminal Examination - 2079

Class : XII

Subject: Mathematics (0081)

Full Marks: 75

Time: 3 hrs.

SET 'A'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

- $C(8, r) - C(7, 3) = C(7, 2)$ if $r =$
 a) 3 b) 4 c) 5 d) 3 or 5
- Which term is free from x in the expansion of $\left(2x + \frac{1}{x}\right)^{10}$?
 a) t_4 b) t_5 c) t_6 d) None of the terms
- $\left(\frac{-1+\sqrt{-3}}{2}\right)^4 + \left(\frac{-1-\sqrt{-3}}{2}\right)^4 =$
 a) -1 b) 0 c) 1 d) 2
- $\cos(\tan^{-1} x) =$
 a) $\sqrt{1+x^2}$ b) $\frac{1}{\sqrt{1+x^2}}$ c) $1+x^2$ d) $\sqrt{1-x^2}$
- A conic has an eccentricity less than unity. Then it is
 a) an ellipse b) a hyperbola c) a parabola d) a circle
- The distance of the point $(4, 3, 5)$ from the y -axis is
 a) $\sqrt{34}$ b) 5 c) $\sqrt{41}$ d) $\sqrt{15}$

7. For two vectors $\vec{a}$ and $\vec{b}$, $|\vec{a}|=3$, $|\vec{b}|=\frac{\sqrt{2}}{3}$ and $\vec{a} \times \vec{b}$ is a unit vector, then angle between $\vec{a}$ and $\vec{b}$ is
- a) $\frac{\pi}{4}$ b) $\frac{\pi}{6}$ c) $\frac{\pi}{2}$ d) $\frac{\pi}{3}$
8. The binomial distribution whose mean is 9 and standard deviation $\frac{3}{2}$ is
- a) $\left(\frac{3}{4} + \frac{1}{4}\right)^{12}$ b) $\left(\frac{1}{3} + \frac{2}{3}\right)^{12}$
- c) $\left(\frac{1}{4} + \frac{3}{4}\right)^{12}$ d) both (a) and (c)
9. If $y = \frac{e^x - e^{-x}}{2}$ then $\frac{dy}{dx} =$
- a) $\sinh x$ b) $\cosh x$ c) $e^x + e^{-x}$ d) $-\cosh x$
10. $\int \frac{dx}{\sqrt{25-x^2}} dx =$
- a) $\sin^{-1}x + c$ b) $\sin^{-1}\left(\frac{x}{5}\right) + c$
- c) $\cos^{-1}\left(\frac{x}{5}\right) + c$ d) $\ln(x + \sqrt{25-x^2}) + c$
11. During the process of solving system of linear equations in x, y, z , we obtain $0 \times z = 0$ at the end of forward elimination. The system is
- a) inconsistent and has no solution.
- b) consistent and has unique solution.
- c) consistent and has only trivial solution.
- d) consistent and has infinitely many solutions.

Group 'B'

Short Answer Questions.

[8 × 5 = 40]

12. a) In how many ways can the letters of the word “ANSWER” be arranged? In how many ways can it be done if the relative positions of vowels and consonants are not changed? [1+2]

- b) From 6 gentlemen and 4 ladies, a committee of 5 is to be formed. In how many ways can this be done so as to include at least 3 ladies? [2]
13. a) If $(1 + x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$, prove that:
- $$C_0C_n + C_1C_{n-1} + C_2C_{n-2} + \dots + C_nC_0 = \frac{(2n)!}{n!n!}. \quad [3]$$
- b) If $y = \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ to ∞ , $|y| < 1$, then prove that:
- $$x = y - \frac{y^2}{2} + \frac{y^3}{3} - \frac{y^4}{4} + \dots \text{ to } \infty. \quad [2]$$
14. a) Find the equation of hyperbola in standard position with focus at $(6, 0)$ and a vertex at $(4, 0)$. [2]
- b) Find the eccentricity and foci of the ellipse $9x^2 + 5y^2 - 30y = 0$. [3]
15. a) Prove that: $\cos^{-1} x = 2 \sin^{-1} \sqrt{\frac{1-x}{2}}$ [2]
- b) If $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \pi$, show that $x + y + z = xyz$. [3]
16. a) If α, β and γ are the direction angles of a line, prove that $\cos 2\alpha + \cos 2\beta + \cos 2\gamma + 1 = 0$. [2]
- b) Solve using inverse matrix method.
- $$4x + 5y = 2, \quad 2x + 3y = 0 \quad [3]$$
17. a) Find a unit vector perpendicular to the plane containing the vectors $\vec{i} + \vec{j} - 2\vec{k}$ and $\vec{i} - 2\vec{j} + \vec{k}$. [2]
- b) In any triangle ABC , prove vectorically that
- $$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c} \quad [3]$$
18. a) Integrate: $\int \frac{dx}{x^2 - 4x + 5}$ [2]
- b) Prove that: $\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2}) + c$ [3]
19. Using simplex method, maximize $Z = 25x + 45y$ subject to $x + 3y \leq 21, 2x + 3y \leq 24, x, y \geq 0$. [5]

Group 'C'

Long Answer Questions.

[3 × 8 = 24]

20. a) A class consist of 60 boys and 40 girls. If two students are chosen at random, what is the probability that
- (i) both are boys ?
 - (ii) both are girls ?
 - (iii) one is boy and one is girl ? [3]
- b) The average percentage of failure in a certain examination is 70. What is the probability that out of 5 candidates, at least 3 will pass? [2]
- c) Solve the following system of equations by using Gauss–Elimination method: $3x - 2y + 2z = 1$, $2y + 3z = 7$, $2x + 3y - 3z = 5$ [3]
21. a) State De-Moivre's theorem. Apply it to find the square roots of $4 + 4\sqrt{3}i$. [5]
- b) Find the equation of the plane through the points $(-1, 1, 1)$ and $(1, -1, 1)$ and perpendicular to the plane $x + 2y + 2z = 5$. [3]
22. a) Show that the tangents to the curve $y = x^2 - 5x + 6$ at the points $(2, 0)$ and $(3, 0)$ are right angles. [2]
- b) Using L Hospital's rule, evaluate: $\lim_{x \rightarrow 0} \frac{\log (\sin x)}{\cot x}$. [3]
- c) State Mean Value Theorem. Mention an additional condition that will satisfy the Rolle's Theorem. Rolle's Theorem is not applicable for the function $f(x) = \frac{1}{x^2}$ in the interval $[-1, 1]$, why? [1+1+1]

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